Test Intelligent Systems NU-5A91

Number of questions: 31
Session period: 22 december 2023 08:30 - 11:45
Duration: —

Instruction

Exam duration: 2 HOURS 15 Minutes, unless you get 30 minutes extra time.

The following tools are permitted:
- Calculator
- Scrap paper (nothing written or printed on it)
- Formula sheet intelligent systems 2nd partial

This second partial exam consists of 31 questions of varying complexity. In total you can reach maximally: 120 Points

There are 7 topics according to the main topics of lectures 8-13: Uncertainty & Bayesian networks, ML 1, ML 2, ML 3, and ML 4. Make sure you do not lose too much time on individual questions.

Good luck!

Multiple *Choice* questions have only **one** correct solution. In this case, the options are given by circles. **Multiple *Answers*** questions have one or more correct solution. In this case, the options are given by squares. The score of Multiple Answer questions is calculated automatically by the system as follows: Score = proportion good * (1 - proportion wrong * (1/a + (((n-k)/n)*(1-1/a))), where n is the total number of alternatives, k the number of correct answers and a a weighting factor set to 2. The more correct answers you give the more points you get, but wrong answers also count negatively.

Vagueness, Uncertainty & Bayesian Networks

The following set of questions are about uncertainty and Bayesian networks.

There are 5 questions for a total of 13 points.

Question 1 – Multiple choice – Question-ID: 205064 (2 points)

Consider the statement: “All elephants have some greyish color”. Which of the following statements is true?

A The author of this statement describes the color of elephants with a degree of vagueness.

B The author of this statement describes the color of elephants with a degree of uncertainty.

Question 2 – Fill in (numerical) – Question-ID: 404038 (6 points)

In Fuzzy Logic, the meaning of a formula is defined as a value between 0 and 1. What would be the meaning of f(((Q & P) v P) -> (¬Q v ¬P)) if f(P)=0.7 and f(Q)=0.4?

Provide a number rounded to **one** digit after the decimal point. Use dot as a separator (e.g. 0.22).

Question 3 – Fill in (multiple) – Question-ID: 405474 (6 points)

Consider the following relations, where WGS-5 and WGS-6 stands for Working Group Session 5 and Working Group Session 6, respectively.

Understanding:

	WGS5	WGS6
Joana	0.3	0.7
Andreea	0.6	0.4
Francesca	0.8	0.2

Contains;

	Fuzzy Logic	Probability
WGS-5	0.7	0.3
WGS-6	0.4	0.6

Fill in the table for the composition of the two relations.

	Fuzzy Logic	**Probability**
Joana		
Andreea		
Francesca		

Question 4 – Multiple choice – Question-ID: 404033 (5 points)

Consider the following joint probability distribution table.

	headache		¬ headache	
	fever	¬ fever	fever	¬ fever
cold	0.258	0.094	0.135	0.015
¬ cold	0.012	0.063	0.036	0.387

What is the probability of **P**(Fever | headache)?

A < 0.37, 0.63 >

B < 0.78, 0.22 >

C < 0.30, 0.70 >

D < 0.63, 0.37 >

Question 5 – Multiple choice – Question-ID: 404034 (3 points)

Two fair dice are thrown, and the numbers uppermost are observed. Decide whether the following events are dependent or independent and what are their probabilities:

A: The difference between the numbers on the two dice is 3.
B: The number on the red die is odd.

A Dependent, P(A) = 1/6 , P(A|B) = 3/9

B Independent, P(B) = 6/36, P(B|A) = 1/6

C Independent, P(A) = 6/36, P(A|B) = 1/6

D Dependent, P(B) = 12/36, P(B|A) = 1/3

Question 6 – Fill in (multiple) – Question-ID: 404035 (8 points)

Given the following information:

- 219 out of 365 days in Amsterdam are rainy.
- Four out of five days, Anne travels with public transport when it is raining.
- Anne travels with public transport 64% of the time regardless of the weather.

For the following questions, provide a number rounded to **two** digits after the decimal point. Use dot as a separator (e.g. 0.22).

What is the probability of a rainy day in Amsterdam?

What is the probability that Anne is traveling with public transport given that it is raining?

What is the probability that Anne is traveling with public transport regardless of the weather?

What is the probability that today is a rainy day given that Anne has traveled with public transport?

Question 7 – Fill in (multiple) – Question-ID: 292213 (5 points)

Given the following Bayesian network from the previous question, Please fill in the blank:

B	P(B)
+b	0.001
-b	0.999

E	P(E)
+e	0.002
-e	0.998

B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

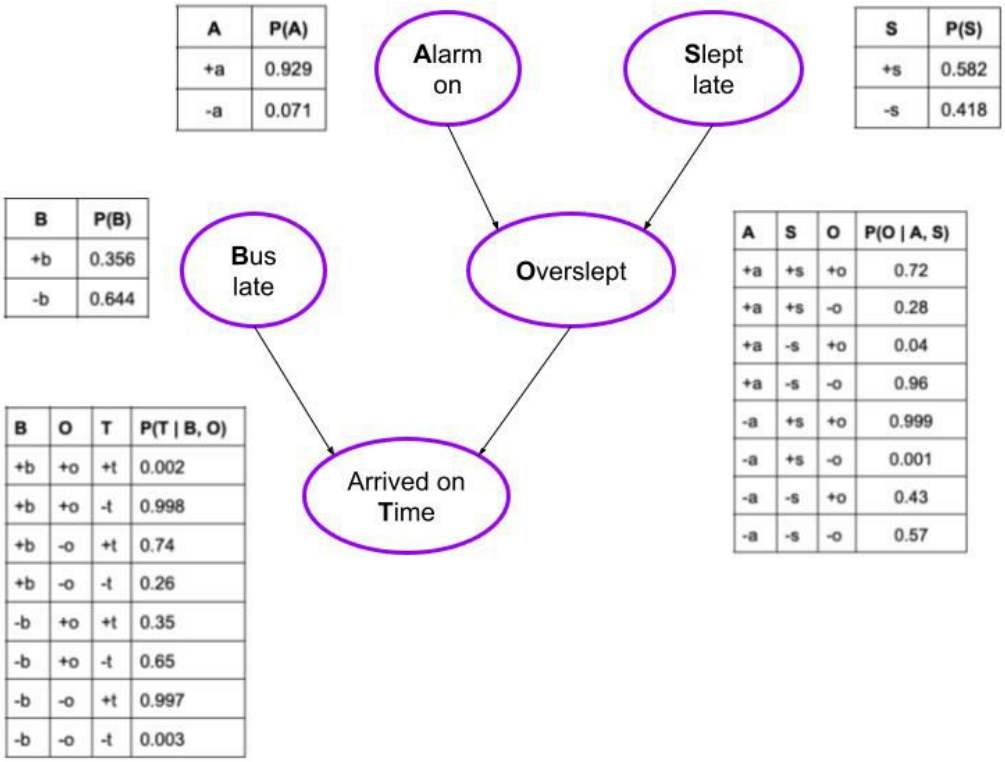
This question requires some more calculation. Make sure you do not loose too much time on it.

What is the probability that the alarm has been activated, but neither burglary nor earthquake has occurred, and both John and Mary call?

(Hint: The full joint distribution is the product of the local conditional distributions $P(J \wedge M \wedge A \wedge \neg B \wedge \neg E)$)

Question 8 – Fill in (multiple) – Question-ID: 404043 (7 points)

Consider the following Bayesian network and the accompanying probability tables about the high school student Tom.



A	P(A)
+a	0.929
-a	0.071

S	P(S)
+s	0.582
-s	0.418

B	P(B)
+b	0.356
-b	0.644

A	S	O	P(O | A, S)
+a	+s	+o	0.72
+a	+s	-o	0.28
+a	-s	+o	0.04
+a	-s	-o	0.96
-a	+s	+o	0.999
-a	+s	-o	0.001
-a	-s	+o	0.43
-a	-s	-o	0.57

B	O	T	P(T | B, O)
+b	+o	+t	0.002
+b	+o	-t	0.998
+b	-o	+t	0.74
+b	-o	-t	0.26
-b	+o	+t	0.35
-b	+o	-t	0.65
-b	-o	+t	0.997
-b	-o	-t	0.003

Provide numbers rounded to **three** digits after the decimal point. Only round at the very end of your calculations. Use dot as a separator (e.g. 0.22).

What is the probability that Tom arrives on time for his classes?

Machine Learning

The following set of questions are about the Machine Learning 1 lecture.

There are 4 questions for a total of 9 points.

Question 9 – Multiple choice – Question-ID: 404400 (2 points)

What is the main difference between regression and classification in machine learning?

A Both regression and classification predict continuous values.

B Classification predicts continuous values, while regression predicts categories.

C Regression predicts continuous values, while classification predicts categories.

D Both regression and classification predict categories.

Question 10 – Multiple choice – Question-ID: 404397 (2 points)

What is the primary purpose of cross-validation in machine learning?

- **A** Testing the model on an unseen dataset
- **B** Training the model on multiple subsets of the dataset
- **C** Fine-tuning hyperparameters using the test set
- **D** Getting a reliable estimate of how well the model generalizes to unseen data

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Question 11 – Multiple choice – Question-ID: 404048 (2 points)

The assistant of a famous photographer has been tasked with ordering the pictures taken by the photograph in different folders based on similarities. However, there is an enormous amount of pictures, so the assistant wishes to take a short-cut instead of sorting the pictures by hand.

What type of machine learning algorithm is most appropriate for this problem?

- **A** Regression
- **B** Classification
- **C** Clustering
- **D** Ranking
- **E** Collaborative filtering

Question 12 – Fill in (multiple) – Question-ID: 291569 (6 points)

What kind of model do we need for the following cases (**classification, regression, ranking, collaborative filtering, clustering**)?
First, indicate whether this would be **supervised or unsupervised** machine learning.

Example:
a) Searching for overlaps in the parts of the population affected by diabetes and arthritis.

unsupervised

clustering

a) Deciding if an MRI image shows a cancerous tumour or not.

b) Twitter is a social networking site that allows micro-blogging services where people broadcast short, public messages refer as tweets. Twitter wants to find out about similar users.

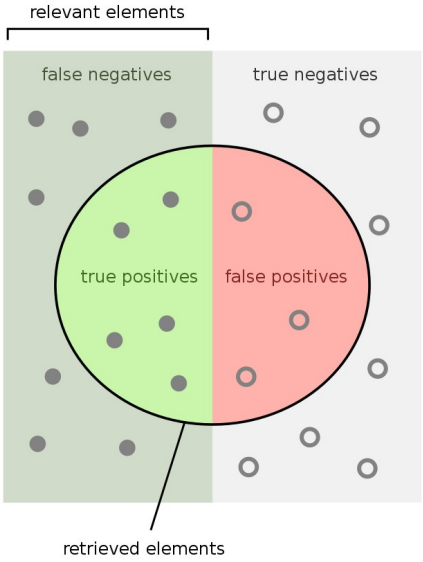
c) Model to forecast the sales number based on the history of the sales.

d) Auto-categorizing the news provided by the News channel.

e) Suggesting a meal in the cafeteria self-order counter based on ratings of meals given by patients and hospital workers.

Question 13 – Multiple response (limited) – Question-ID: 404129 (2 points)

Consider the following diagram and choose the right definitions.



(max. 2 answers)

A Recall refers to how many relevant items are retrieved.

B Recall refers to how many retrieved items are relevant.

C Precision refers to how many relevant items are retrieved.

D Precision refers to how many retrieved items are relevant.

Question 14 – Fill in (multiple) – Question-ID: 404053 (4 points)

Calculate precision and recall for the following system, which classifies pictures of insects as butterflies and NOT butterflies.

Picture	Predicted Class	Actual Class
	Butterfly	Butterfly
	NOT Butterfly	Butterfly
	Butterfly	NOT Butterfly
	Butterfly	Butterfly
	Butterfly	Butterfly
	NOT Butterfly	NOT Butterfly
	NOT Butterfly	Butterfly

Provide numbers rounded to **two** digits after the decimal point. Use dot as a separator (e.g. 0.22).

Precision:

Recall:

Question 15 – Multiple choice – Question-ID: 404058 (2 points)

Consider the following output of an Ranking algorithm created to retrieve movies with Keira Knightley:

1. Pirates of the Caribbean: Dead Man's Chest*
2. Barbie
3. The Imitation Game*
4. Pirates of the Caribbean: At World's End*
5. Doctor Strange
6. Pride & Prejudice*
7. Dune
8. Official Secrets*
9. Atonement*
10. A Dangerous Method*
11. The Duchess*
12. Spider-Man: Homecoming

Only the titles with a star (*) at the end represent movies in which Keira Knightley has **actually** starred.

What is the precision at 9 and the precision at 11 of this ranking?

A 6/9, 7/11

B 6/9, 8/11

C 8/9, 8/11

D 8/11, 6/9

Question 16 – Multiple choice – Question-ID: 404403 (2 points)

What is a characteristic of Naive Bayes classifiers?

A They are primarily used for regression tasks.

B They assume independence between features.

C They are prone to overfitting.

D They are sensitive to irrelevant features.

Question 17 – Fill in (multiple) – Question-ID: 404050 (7 points)

Consider the following list of feature vectors, obtained from five documents about butterflies (B) and about moths (M).

	wings	**head**	**threadlike**	**legs**	**camouflage**	**antennae**	CLASS
1	0	0	0	1	1	1	B
2	1	0	1	0	1	0	M
3	0	1	0	1	0	1	B
4	1	0	0	0	0	0	B
5	0	1	0	0	1	1	M

Using the Boolean features **{wings, head, threadlike, legs, camouflage, antennae}**, make a Naive Bayes Classifier to classify the following document as belonging to either the Butterfly or Moth class.

		CLASS
6	The antennae on their head are composed of many segments and have threadlike tips**.**	?

What is the probability that the new document is from class M? Provide a number rounded to **five** digits after the decimal point. Use dot as a separator (e.g. 0.22).

What is the probability that the new document is from class B? Provide a number rounded to **six** digits after the decimal point. Use dot as a separator (e.g. 0.22).

Based on the above probabilities, therefore, the document should be classified as:

Question 18 – Fill in (multiple) – Question-ID: 404584 (5 points)

Convert each of these sentences into feature vector after removing punctuations, stemming and stop-words (CHECK: **only verbs, adjectives and nouns are NOT stopwords in this example)**.

a. Dog walks in the field.
b. The cow sleeps on the poppies.
c. The cats and the dogs run in the field.
d. The cow is mad because the cats and the dogs walked all over the poppies.

Order the features in the order in which they appear in the sentences from left to right. The first row is for the words.

Machine Learning 2

The following set of questions are about the Machine Learning 2 lecture.

There are 5 questions for a total of 13 points.

Question 19 – Multiple choice – Question-ID: 404404 (2 points)

How does k-means clustering work?

A It uses a probability distribution to cluster data points.

B It assigns each data point to the nearest centroid.

C It calculates the mean of each feature in the dataset.

D It performs a linear transformation on the input features.

Question 20 – Multiple choice – Question-ID: 404130 (2 points)

Which of the following will be Euclidean Distance between the two data points (4,6) and (6,4)?

A √4

B -4

C 2

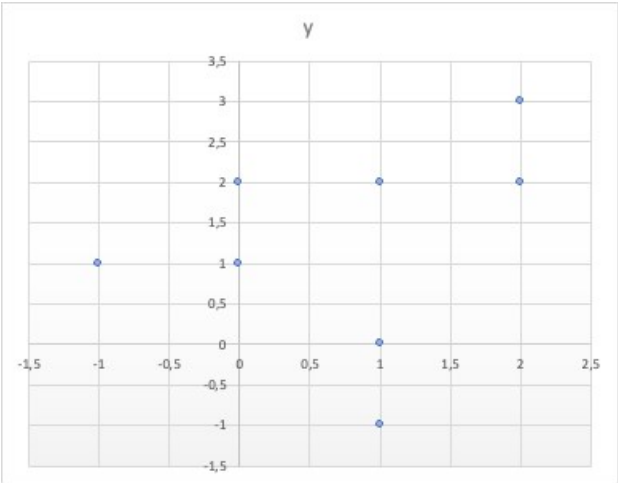
D √8

Question 21 – Multiple choice – Question-ID: 404131 (3 points)

Consider the following data where x and y are 2 input variables and Class is a dependent variable.

x	y	Class
-1	1	-
0	1	+
0	2	-
1	-1	-
1	0	+
1	2	+
2	2	-
2	3	+

Below is the scatter plot of the above data.



Q. What would be the class of data point (1.5, 2.5) be using euclidean distance where k = 3 and k-nearest neighbour?

A -Class

B None of the given options

C +Class

Question 22 – Match (even) – Question-ID: 404483 (3 points)

Two texts are similar when their feature vectors are similar. Which of the following corresponds to which?

A Hamming Distance

B Euclidean Distance

C Cosine Similarity

1 (number of different bit)/(total number of bits)

2 $$\frac{\mathbf{A}\cdot\mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|}=\frac{\sum_{i=1}^{n}A_iB_i}{\sqrt{\sum_{i=1}^{n}A_i^2}\sqrt{\sum_{i=1}^{n}B_i^2}},$$

3 $$\sqrt{\sum_{i=1}^{n}(q_i-p_i)^2}$$

Question 23 – Multiple choice – Question-ID: 404057 (5 points)

Consider the following list of feature vectors.

A	1	0	1	0	1	1	0	0
B	0	0	0	1	0	0	1	1
C	1	1	1	1	0	0	1	0
D	0	0	1	0	1	1	1	0

Which vector represents the nearest (most similar) neighbour of the below vector if we apply Cosine Similarity?

X	0	1	0	0	0	1	1	0

A D

B B

C C

D A

Machine Learning 3

The following set of questions are about the Machine Learning
There are 4 questions for a total of 7 points.

Question 24 – Multiple choice – Question-ID: 404398 (2 points)

In the context of feature creation, what is a design matrix?

A A matrix used for clustering analysis

B A matrix representing the confusion matrix

C A matrix for storing hyperparameter values

D A matrix of new features generated from existing ones.

Question 25 – Multiple choice – Question-ID: 404393 (3 points)

Which of the following is a common practice in machine learning to prevent overfitting?

A Using as few parameters as possible

B Ignoring the validation set during model training

C Increasing the complexity of the model without constraints

D Using the test set for hyperparameter tuning

Question 26 – Multiple choice – Question-ID: 404481 (5 points)

Consider the following simple linear regression model with the hypothesis h(x)= b*x and three data points:

Data Point 1: (2,5)
Data Point 2: (3,7) and
Data Point 3: (5,11)

Find the value of b.

A 2

B 2.263

C 3.636

D 3.363

Question 27 – Multiple choice – Question-ID: 292316 (8 points)

This exercise is about Regression.

Let us try to fit a linear function to a number of training sets where the slope s is known to be 1.

Suppose we have three pairs of values (1,0), (2,0) and (3,1) for two features. What is the optimal fitting function in this model class (so minimises the error rate)?

A f(x)= x - 5/3

B f(x)= x - 3/5

C There is no optimal fitting function

D f(x)= x + 3/5

E Neither of the functions in the other options is the optimal fitting function, but there exist one.

F f(x)= x + 5/3

Machine Learning 4

The following set of questions are about the Machine Learning 4 lecture.

There are 3 questions for a total of 10 points.

Question 28 – Multiple choice – Question-ID: 404405 (2 points)

What is the purpose of the activation function in a neural network?

A To introduce non-linearity to the network

B To compute the mean of the input features

C To determine the learning rate during training

D To calculate the gradient of the loss function

Question 29 – Multiple response – Question-ID: 404128 (2 points)

Sigmoid and Softmax are both activation functions used in neural networks. Which of the following is true about these functions.

A Sigmoid function is used for multi-class classification

B Softmax function is used for multi-class classification

C Sigmoid function is used for binary classification

D Softmax function is used for binary classification

Question 30 – Multiple choice – Question-ID: 404402 (2 points)

Which optimization technique involves adjusting model parameters to minimize a predefined loss function?

A Analytical optimization

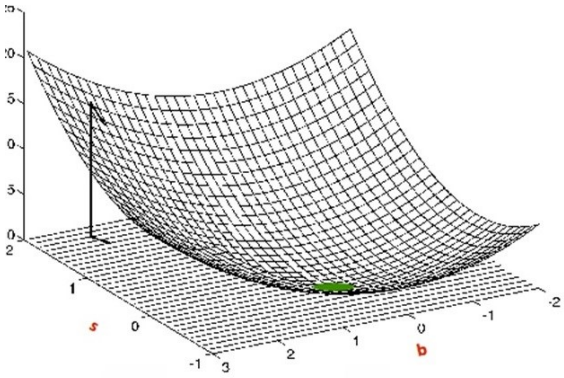
B Gradient-based optimization

C Uninformed optimization

D Hyperparameter tuning

Question 31 – Multiple response – Question-ID: 205100 (5 points)

In the following image, also shown in the lecture, the green point is identified as the optimal solution to a learning problem.



Which of the following statements are true? (multiple answers are possible).

A The third dimension (orthogonal to s and b) shows the error rate for a linear separator function, eg. f(x) = sx+b.

B The third dimension (orthogonal to s and b) enumerates all possible linear separators, so that the green point is the optimal one (as it is the lowest).

C The green point refers to the point in the binary vectorspace with the best training example (the one that fits a particular learning model best)

D The green point refers refers to the linear function with the lowest error rate (the one that fits a particular learning model best according to the training data)

E With a Neural Network we can predict the value of the 3rd dimension (orthogonal to s and b on the x and y axis) given the representation of the features according to variables s and b.

F There are two ways to find the best machine learning model in this scenario: either we analytically find the green point as the point where the derivation of the error function is 0, or we apply gradient descent as a search method to move step by step to the green point. (Gradient descent is a local search similar to hill-climbing)

G The green point is the point where f(x) = sx+b reaches a minimum, ie where the linear separator reaches the lowest point (in other words, f(x)=0

End of test Intelligent Systems NU-5A91